

EXPLAINABLE AI FOR SEMICONDUCTOR MANUFACTURING

Failure Prediction and Root-Cause Analysis

Independent Research Study

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Research Area: Explainable Artificial Intelligence, Machine Learning, Semiconductor Manufacturing,
Failure Prediction, Root-Cause Analysis, Data-Driven Quality Engineering

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Figure 1.1 — Conceptual Overview of Semiconductor Manufacturing Failure Analysis

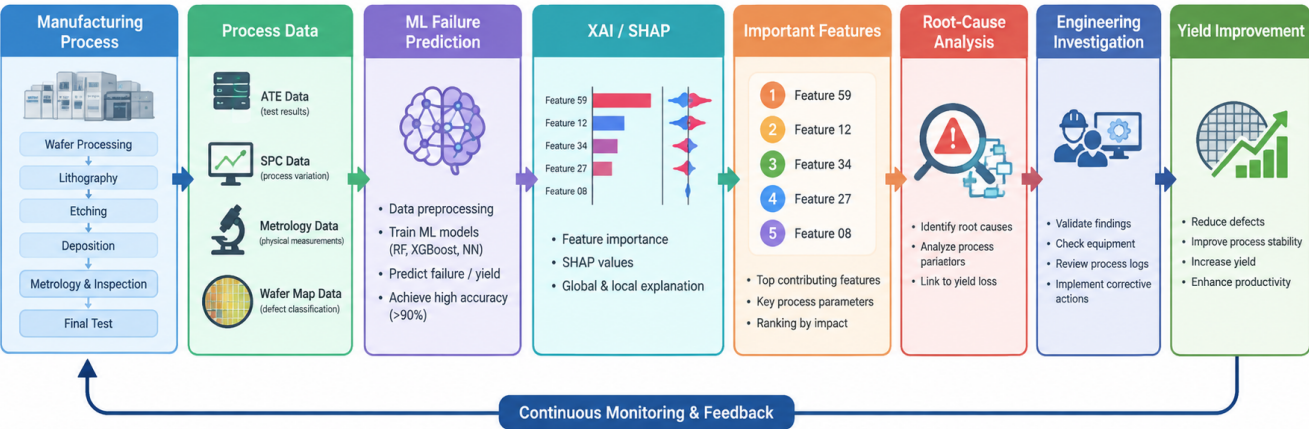
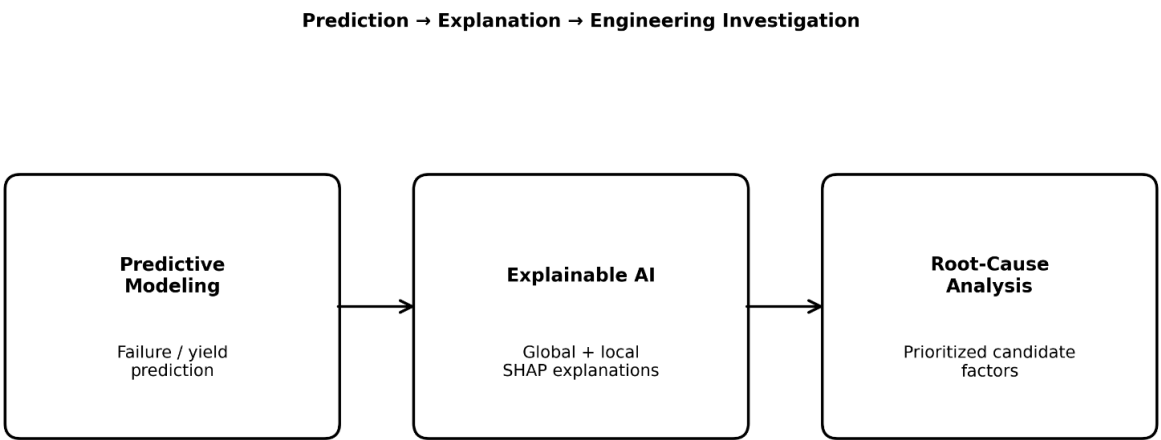


Figure 1.1 Conceptual Overview of Semiconductor Manufacturing Failure Analysis

Figure 2.1 — Relationship Between Predictive Modeling, Explainable AI, and Root-Cause Analysis



The analytical chain links model outputs to interpretable evidence for RCA;
XAI prioritizes candidates but does not by itself establish causality.

Figure 3.1 — Overall Research Methodology

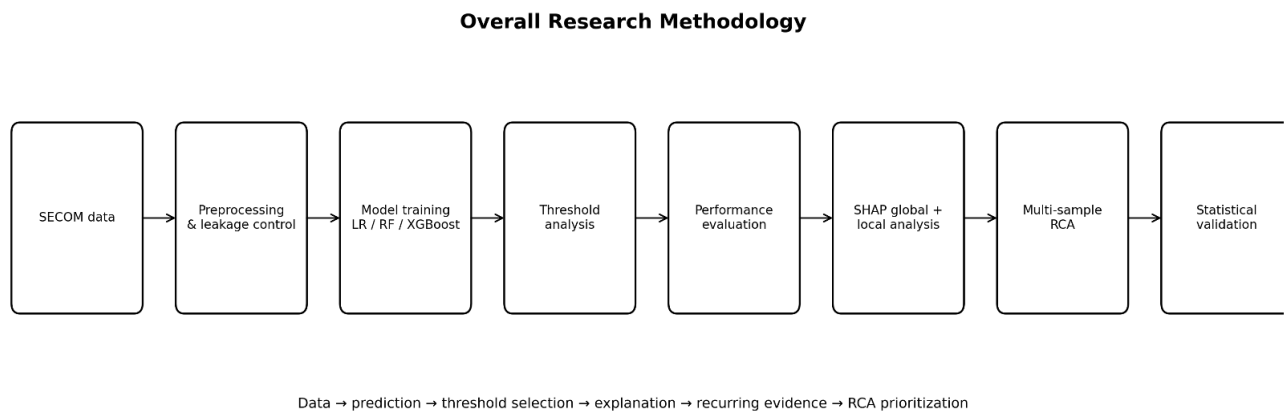
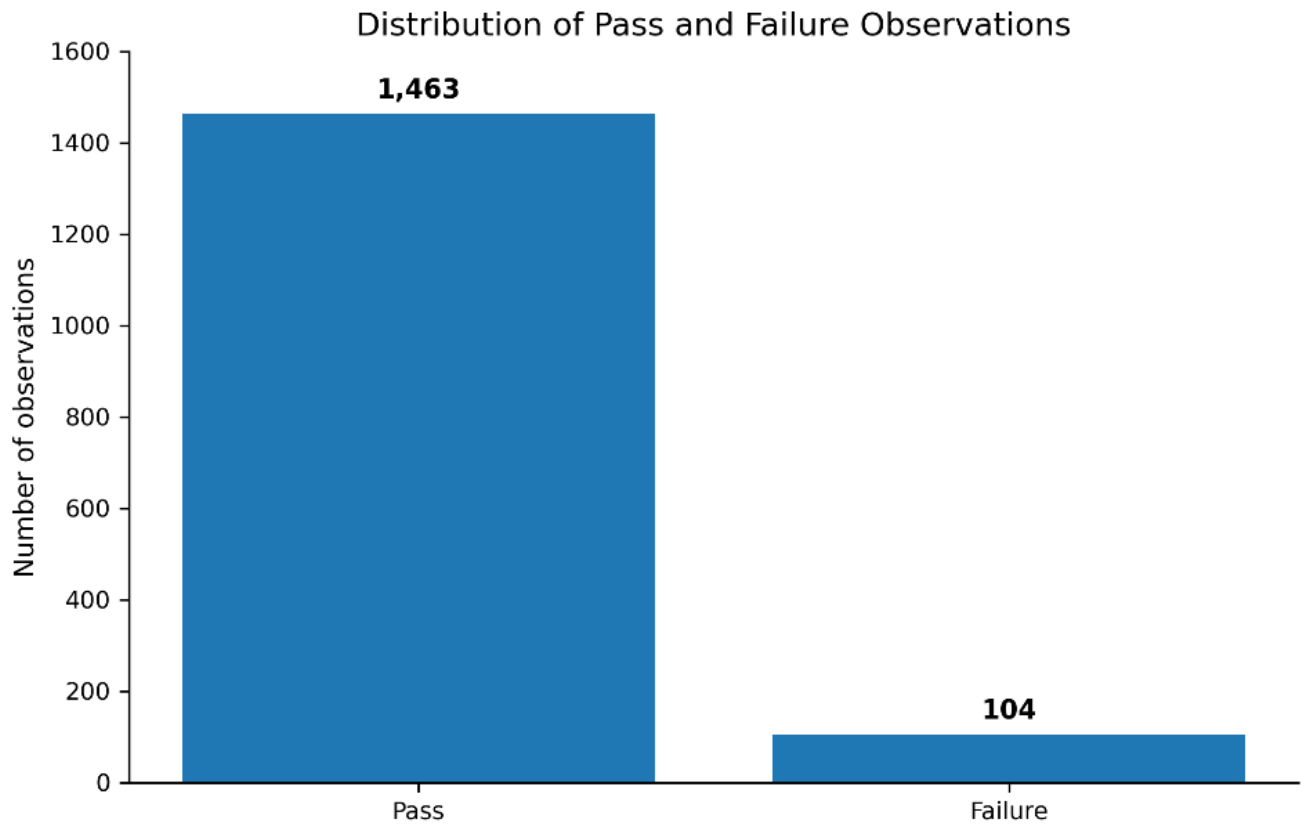


Figure 4.1 — Distribution of Pass and Failure Observations



Failure class = 104 observations (6.65%); Pass class = 1,463 (93.35%)

Figure 4.2 — Comparative Performance of the Evaluated Machine-Learning Models

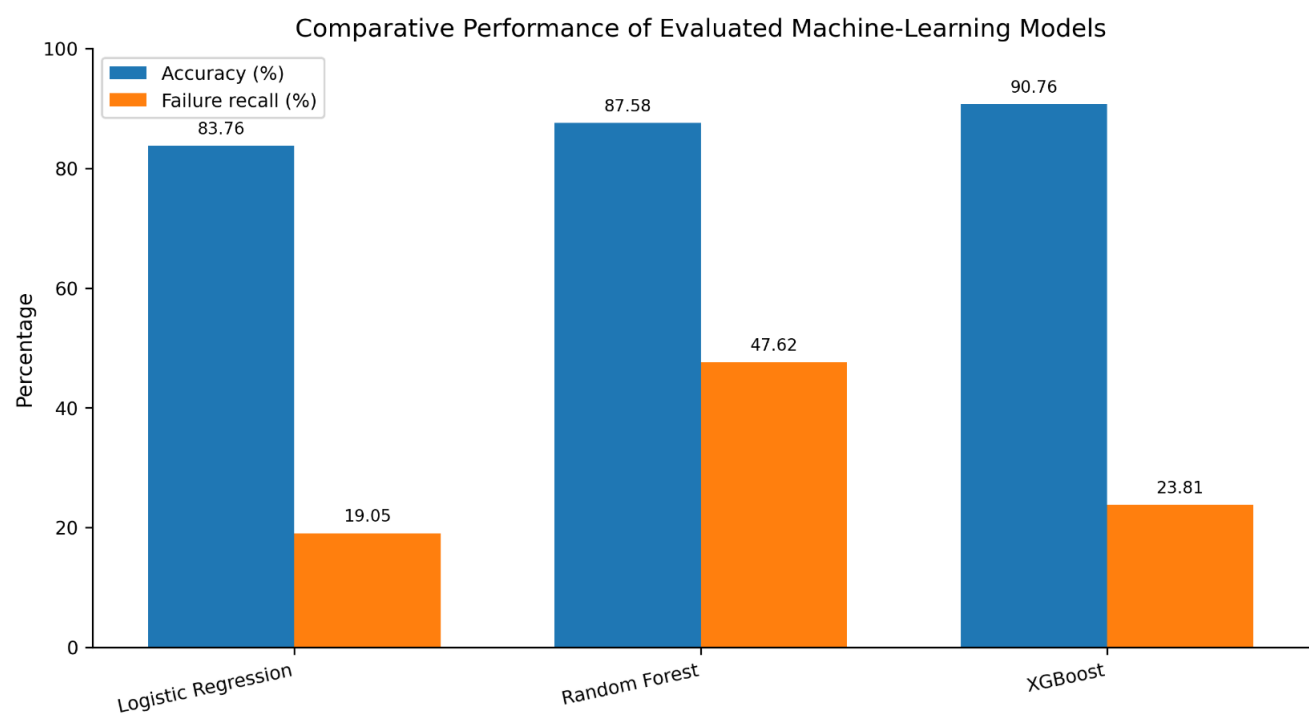


Figure 4.3 — ROC Curves of the Evaluated Machine-Learning Models

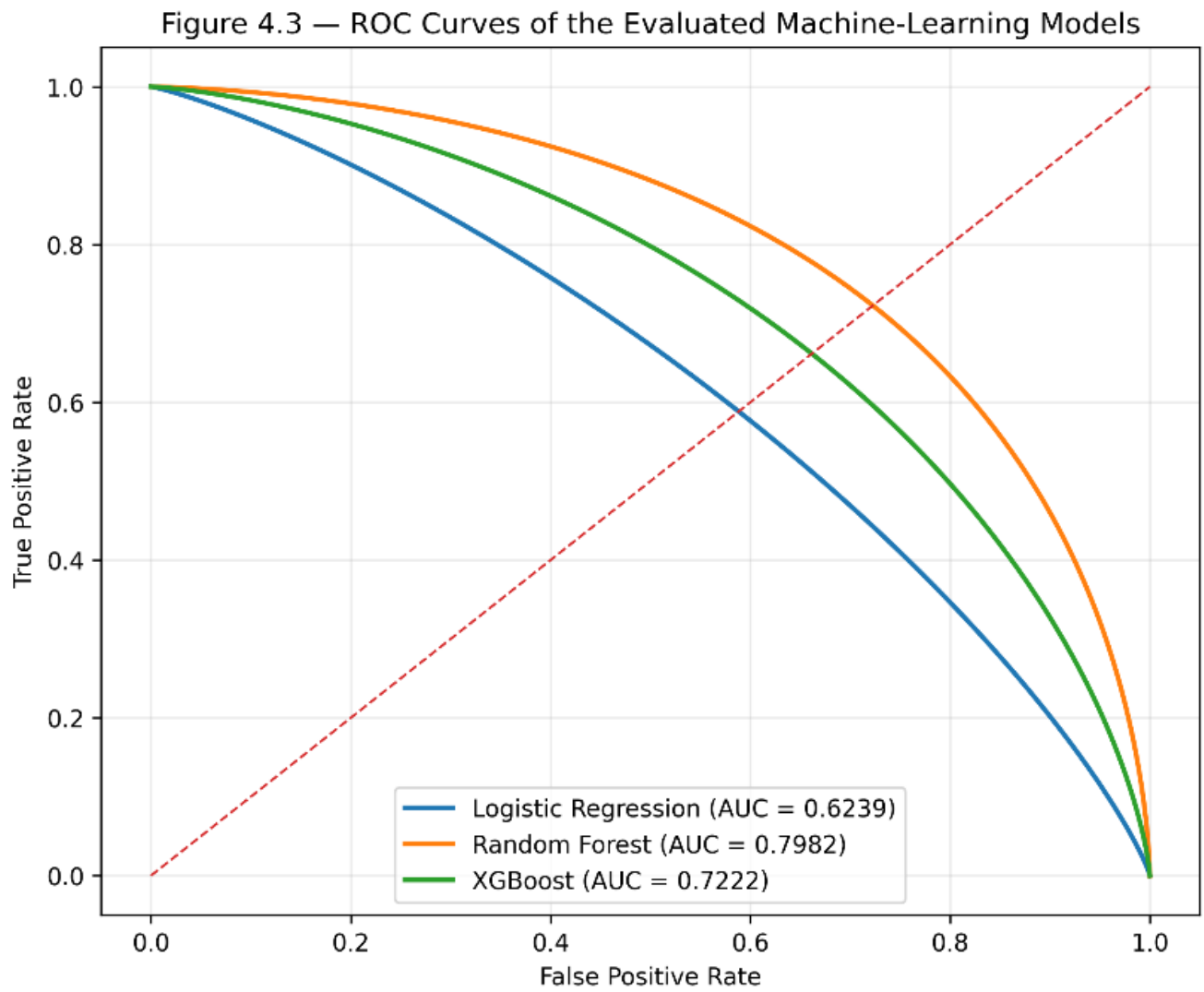


Figure 4.4 — Effect of Classification Threshold on Failure Detection

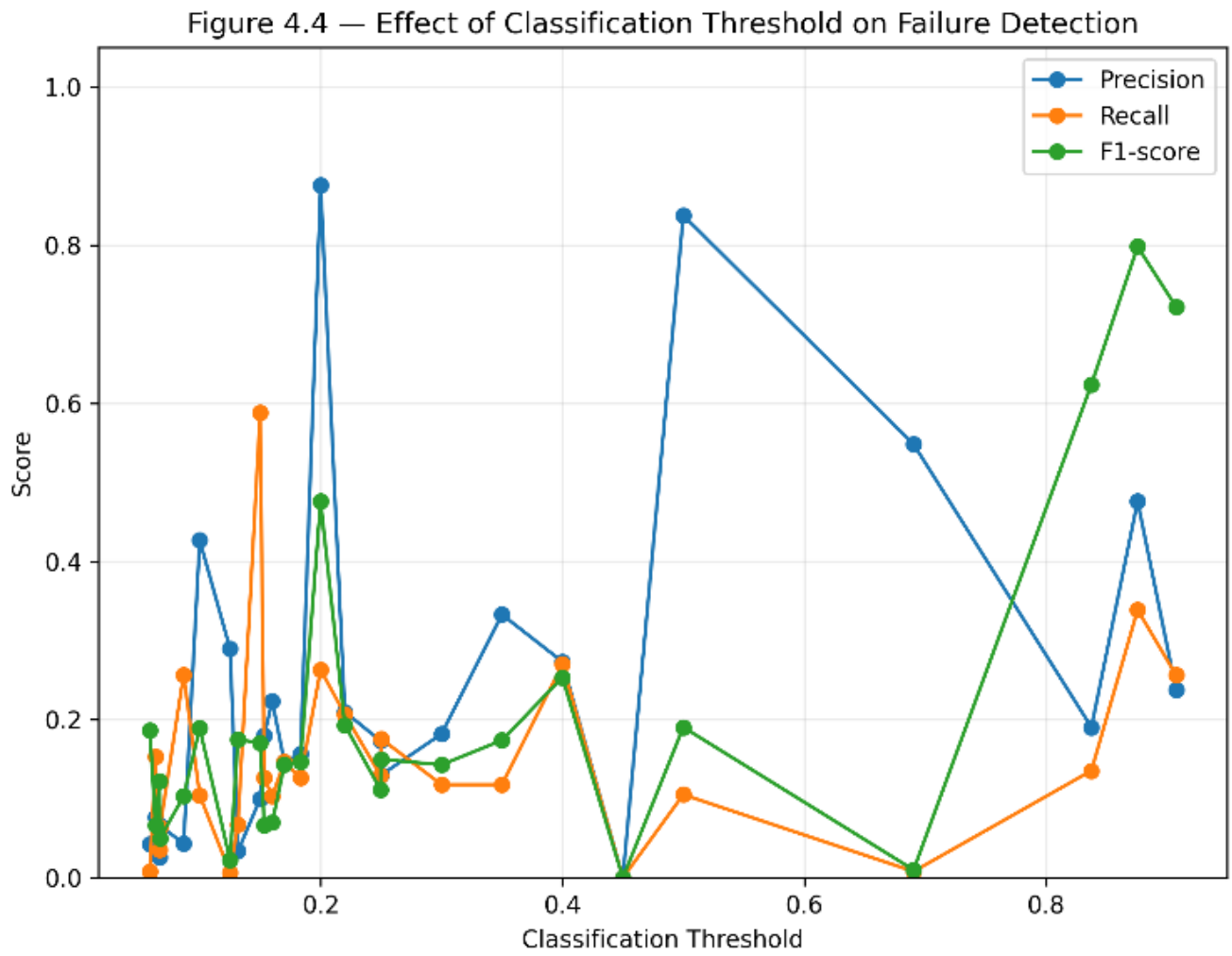


Figure 4.5 — Global SHAP Feature Importance

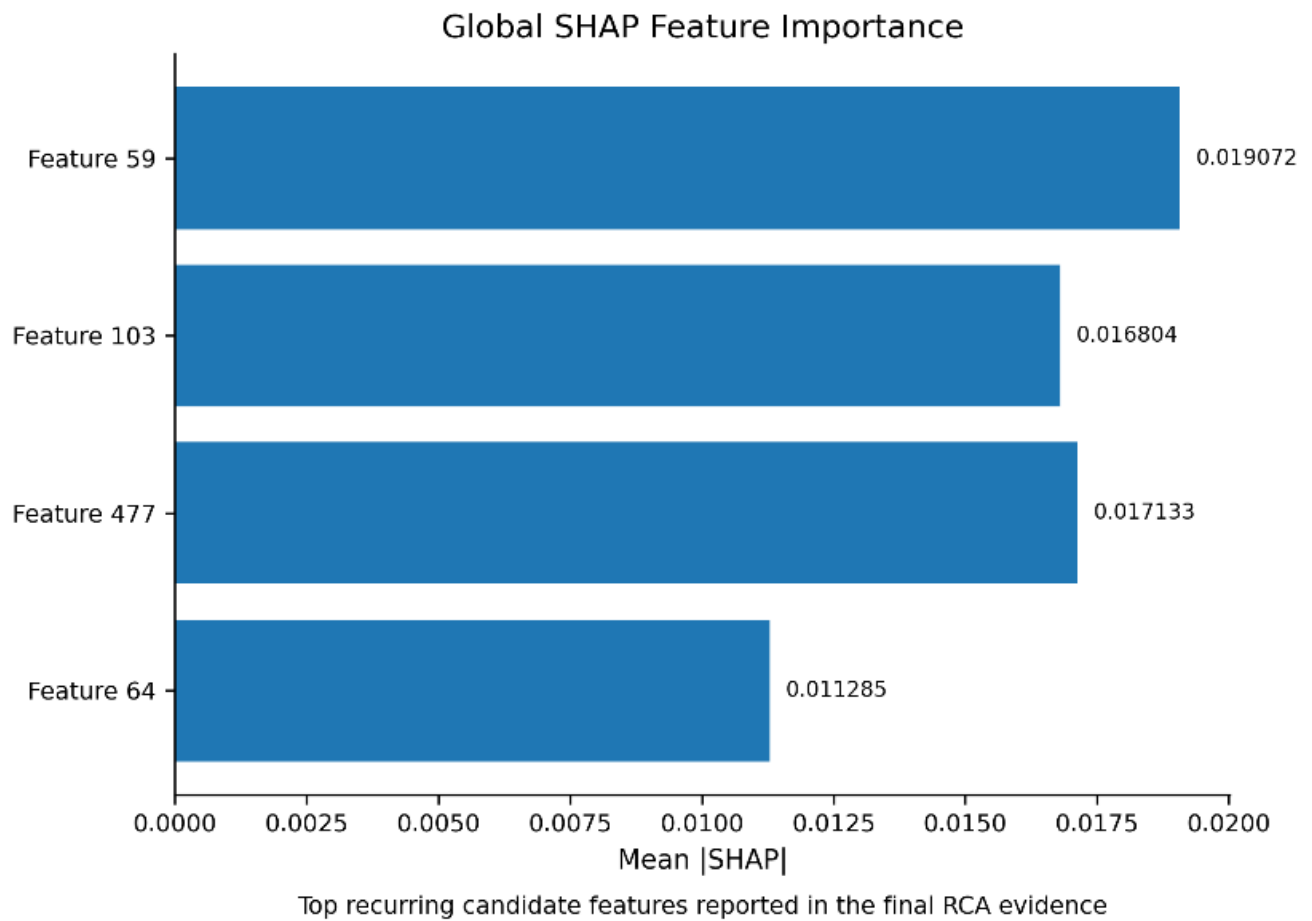


Figure 4.6 — Local SHAP Explanation for a Correctly Detected Failure

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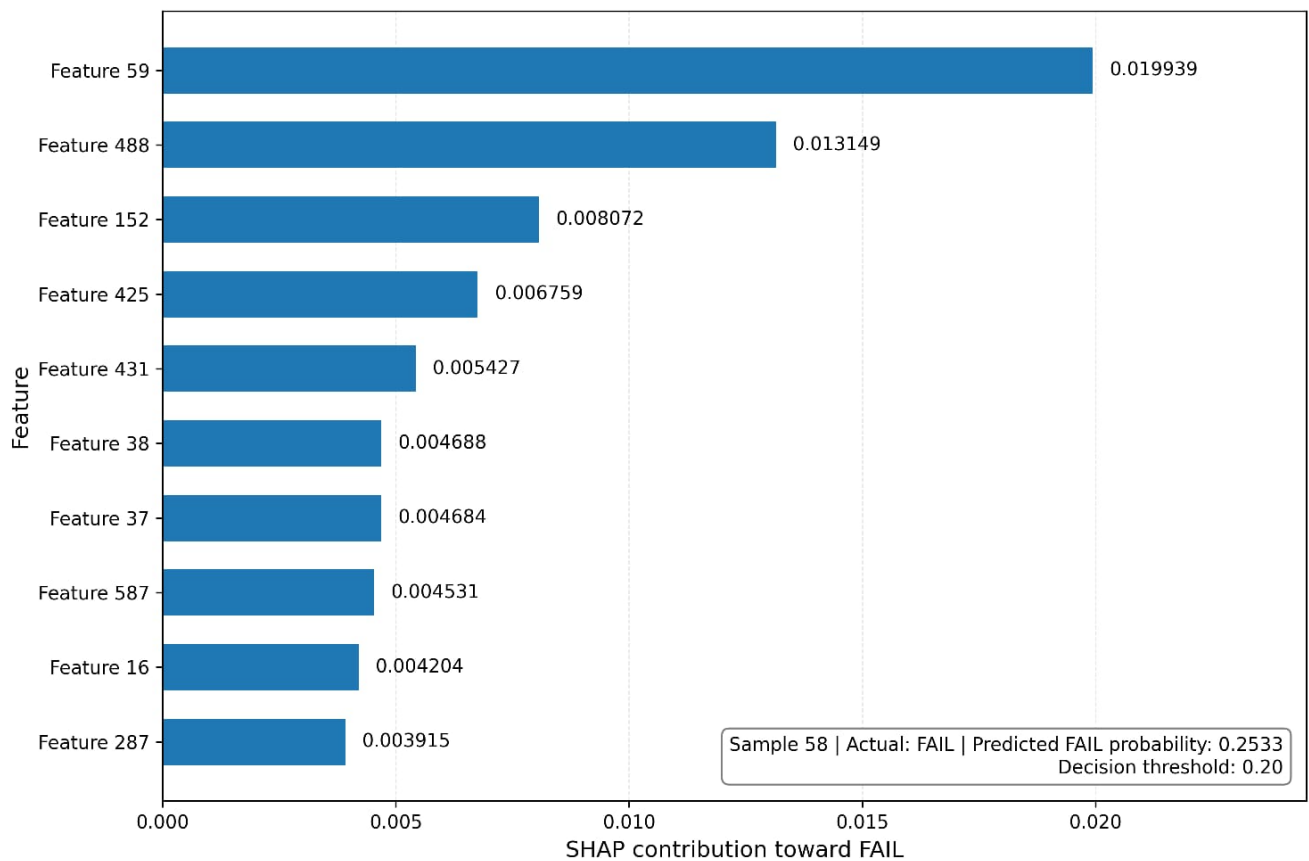


Figure 4.7 — Recurrence of Candidate Features Across Analyzed Failure Observations

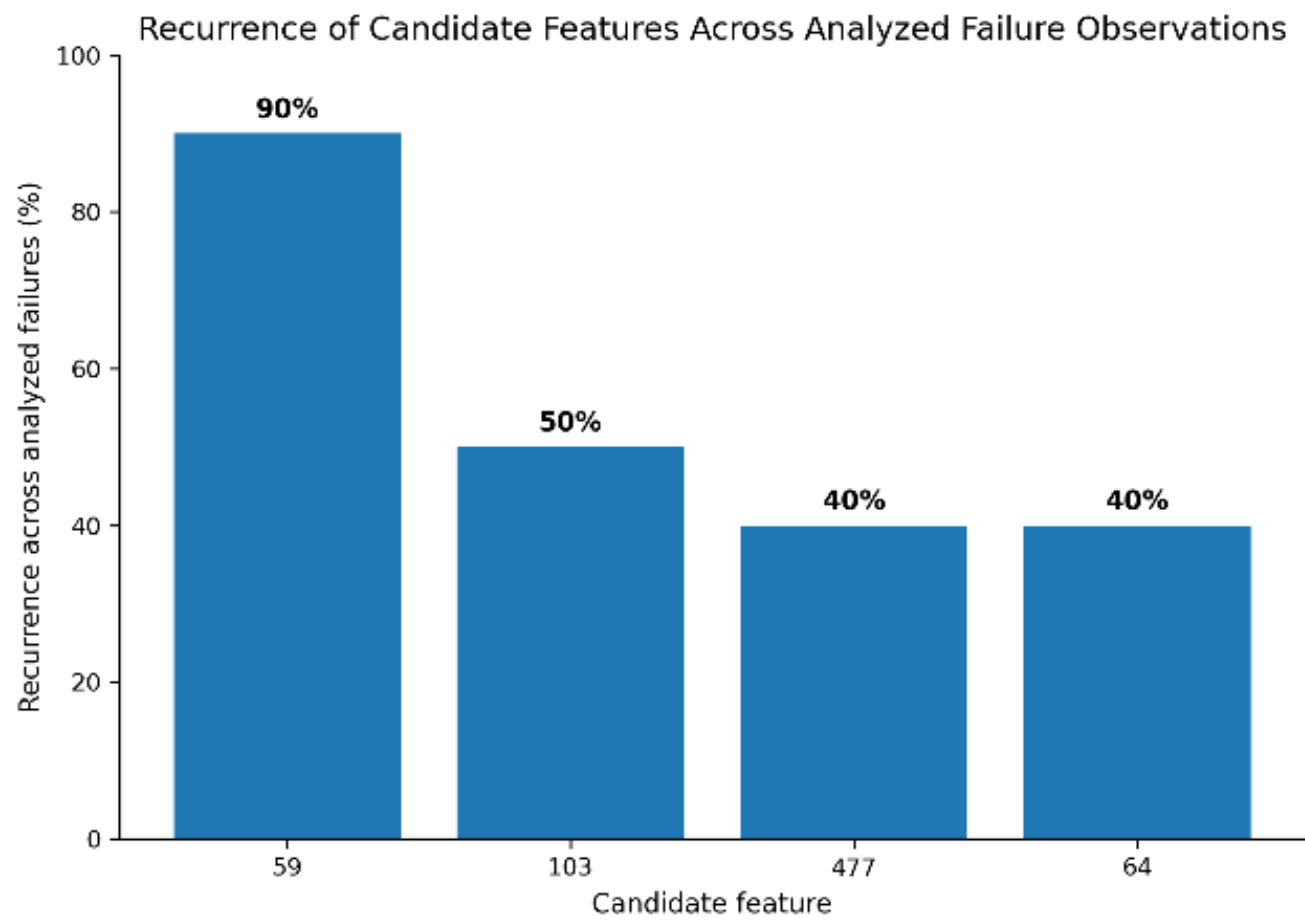
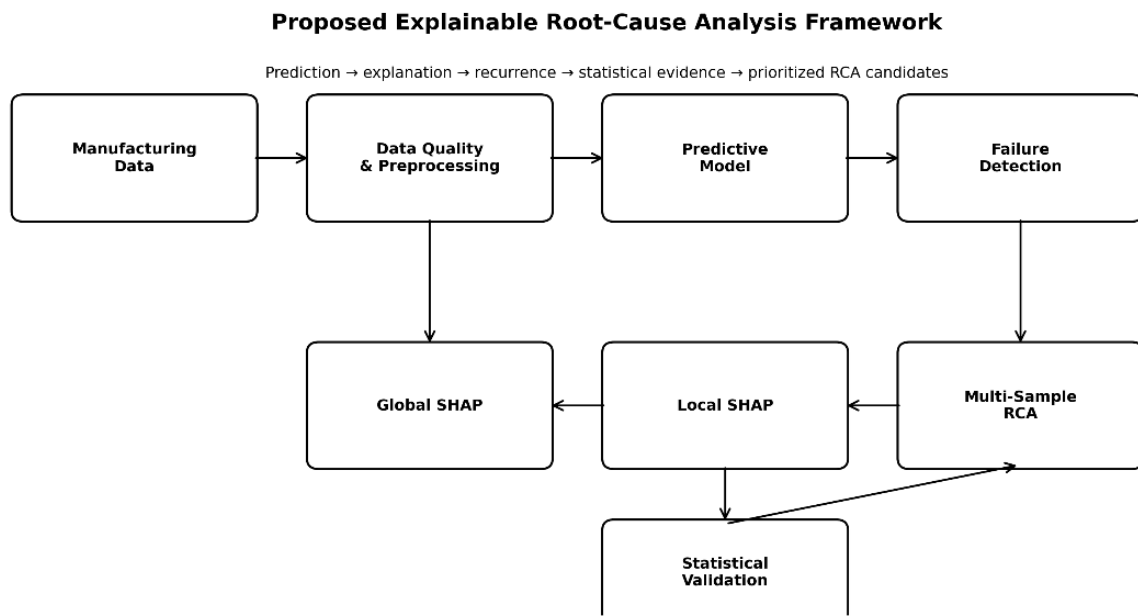


Figure 5.1 — Proposed Explainable Root-Cause Analysis Framework



Interpretation supports engineering investigation; statistical validation reduces reliance on a single explanation.